

# BS-PH101

## Physics-I

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### Combined Module Notes

Modules 1 – 5

Semester I/II • ECE • 4 Credits

CGEC | B.Tech ECE (2023–27)

#### Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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*Exam-Ready Notes*

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# Module 1

## Constraints and Friction

### 1. Constraints and Friction

#### 1.1 Constraints and Degrees of Freedom

##### Definition: Constraint and Degrees of Freedom

A **constraint** is any physical restriction on the position or motion of a mechanical system. The number of independent coordinates required to completely specify the configuration of a system is called its **Degrees of Freedom (DOF)**.

If a system consists of  $N$  particles in 3D space subject to  $k$  independent constraint equations, the number of degrees of freedom is given by:

$$f = 3N - k \quad (1)$$

##### Classification of Constraints:

| Constraint Type      | Condition / Mathematical Form                                                                                | Physical Example                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Holonomic</b>     | Expressible as $f(r_1, r_2, \dots, r_N, t) = 0$ (integrable algebraic relations).                            | Rigid body ( $ \vec{r}_i - \vec{r}_j  = c$ ), simple pendulum ( $r = L$ ).              |
| <b>Non-Holonomic</b> | Expressible as inequalities or non-integrable differential relations ( $f(\vec{r}, \dot{\vec{r}}, t) = 0$ ). | Rolling without slipping on a rough plane, gas molecules in a container ( $r \leq R$ ). |
| <b>Scleronomic</b>   | Constraint equations do not explicitly depend on time $t$ ( $\partial f / \partial t = 0$ ).                 | Bead on a fixed wire ring, rigid dumbbell.                                              |
| <b>Rheonomic</b>     | Constraint equations explicitly depend on time $t$ ( $\partial f / \partial t \neq 0$ ).                     | Bead on a wire ring rotating with constant angular velocity $\omega$ .                  |

#### 1.2 Friction and Laws of Friction

Friction is the resistive contact force that opposes the relative motion or tendency of relative motion between two surfaces in contact.

## Key Laws and Relations of Friction

- **Static Friction ( $f_s$ ):** Self-adjusting force. Maximum value is limiting friction:

$$f_{s,\max} = \mu_s N \quad (2)$$

- **Kinetic Friction ( $f_k$ ):** Constant resistive force during motion:

$$f_k = \mu_k N \quad (\text{where } \mu_k < \mu_s) \quad (3)$$

- **Angle of Friction ( $\lambda$ ):** Angle between total contact reaction  $\vec{R}$  and normal reaction  $\vec{N}$ :

$$\tan \lambda = \frac{f_{s,\max}}{N} = \mu_s \implies \lambda = \tan^{-1}(\mu_s) \quad (4)$$

- **Angle of Repose ( $\theta_r$ ):** Maximum angle of inclination of a rough plane at which a body remains at rest without sliding:

$$mg \sin \theta_r = \mu_s mg \cos \theta_r \implies \tan \theta_r = \mu_s \implies \theta_r = \lambda \quad (5)$$

## Solved Problem 1: Block on an Inclined Plane with Friction

**Problem:** A block of mass  $m = 5$  kg rests on an inclined plane of angle  $\theta = 30^\circ$ . The coefficient of static friction is  $\mu_s = 0.4$ . Determine whether the block slides down, and find the frictional force.

**Solution:**

1. Component of gravity along the incline:  $F_{\text{down}} = mg \sin 30^\circ = 5 \times 9.8 \times 0.5 = 24.5$  N.
2. Normal reaction:  $N = mg \cos 30^\circ = 5 \times 9.8 \times 0.866 = 42.435$  N.
3. Maximum limiting static friction:  $f_{s,\max} = \mu_s N = 0.4 \times 42.435 = 16.97$  N.
4. Since  $F_{\text{down}}(24.5 \text{ N}) > f_{s,\max}(16.97 \text{ N})$ , the block **will slide down**. The friction force acting up the plane during motion is kinetic friction  $f_k = \mu_k N$ .

## Solved Problem 2: Degrees of Freedom of Connected Bodies

**Problem:** Find the degrees of freedom for: (a) A double pendulum moving in a plane, (b) Two particles connected by a rigid rod free to move in 3D space.

**Solution:**

1. **Double Pendulum in 2D:**  $N = 2$  particles, constrained to plane ( $z_1 = 0, z_2 = 0$ ), fixed length  $r_1 = L_1$  and  $|\vec{r}_2 - \vec{r}_1| = L_2$ . Total independent angles  $\theta_1, \theta_2$ . Hence **DOF = 2**.
2. **Rigid Dumbbell in 3D:**  $N = 2$  particles in 3D (6 coordinates) subject to  $k = 1$  constraint ( $|\vec{r}_1 - \vec{r}_2| = d$ ). Hence **DOF = 3(2) - 1 = 5** (3 translational, 2 rotational).

## 2. Vector Calculus and Potential Energy Function

## 2.1 Gradient, Divergence, and Curl

### Definition: Gradient, Divergence, and Curl

- **Gradient** ( $\nabla V$ ): Vector representing the rate and direction of maximum spatial change of a scalar field  $V(x, y, z)$ .
- **Divergence** ( $\nabla \cdot \vec{A}$ ): Scalar representing the net outward flux of a vector field per unit volume.
- **Curl** ( $\nabla \times \vec{A}$ ): Vector representing the circulation or rotational density of a vector field.

### Mathematical Definitions in Cartesian Coordinates

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \quad (6)$$

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad (7)$$

$$\text{curl } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (8)$$

## 2.2 Potential Energy Function $F = -\text{grad } V$

For a conservative force field  $\vec{F}$ , the work done along a closed path is zero ( $\oint \vec{F} \cdot d\vec{r} = 0$ ). This implies  $\vec{F}$  can be expressed as the negative gradient of a scalar potential energy function  $V(\vec{r})$ :

$$\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k}\right) \quad (9)$$

**Physical Meaning of Negative Sign:** The force acts in the direction of *decreasing* potential energy (towards stable equilibrium).

**Equipotential Surfaces:** A surface on which the potential energy  $V(\vec{r}) = \text{constant}$ .

- Work done in moving a mass along an equipotential surface is zero:  $dW = \vec{F} \cdot d\vec{r} = -dV = 0$ .
- The force vector  $\vec{F}$  is everywhere perpendicular to the equipotential surface passing through that point.

### Solved Problem 3: Deriving Force from Potential Energy

**Problem:** Given potential energy  $V(x, y, z) = 3x^2y - y^3z^2$ , calculate force  $\vec{F}$  at point  $(1, -1, 2)$ .

**Solution:**

1. Calculate partial derivatives:  $\frac{\partial V}{\partial x} = 6xy$ ,  $\frac{\partial V}{\partial y} = 3x^2 - 3y^2z^2$ ,  $\frac{\partial V}{\partial z} = -2y^3z$ .
2. Evaluate at  $(1, -1, 2)$ :  $\frac{\partial V}{\partial x} = 6(1)(-1) = -6$ ,  $\frac{\partial V}{\partial y} = 3(1)^2 - 3(-1)^2(2)^2 = 3 - 12 = -9$ ,  $\frac{\partial V}{\partial z} = -2(-1)^3(2) = 4$ .
3. Substitute into  $\vec{F} = -\nabla V$ :

$$\vec{F} = -(-6\hat{i} - 9\hat{j} + 4\hat{k}) = 6\hat{i} + 9\hat{j} - 4\hat{k} \text{ N} \quad (10)$$

### 3. Conservative Forces and Conservation Laws

#### 3.1 Conservative vs. Non-Conservative Forces

Comparative Breakdown:

| Property          | Conservative Force                                      | Non-Conservative Force                         |
|-------------------|---------------------------------------------------------|------------------------------------------------|
| Work Dependence   | Work done depends only on initial and final positions.  | Work done depends on the actual path taken.    |
| Closed Path Work  | $\oint \vec{F} \cdot d\vec{r} = 0$ .                    | $\oint \vec{F} \cdot d\vec{r} \neq 0$ .        |
| Curl Condition    | $\nabla \times \vec{F} = 0$ (Irrotational).             | $\nabla \times \vec{F} \neq 0$ .               |
| Potential Energy  | Expressible as $\vec{F} = -\nabla V$ .                  | Potential energy function cannot be defined.   |
| Mechanical Energy | Total mechanical energy ( $E = K + V$ ) is conserved.   | Mechanical energy is dissipated as heat/sound. |
| Examples          | Gravitational force, Electrostatic force, Spring force. | Friction, Viscosity, Air resistance.           |

#### 3.2 Fundamental Conservation Laws

##### Conservation Theorems

1. **Conservation of Linear Momentum:** If net external force  $\vec{F}_{\text{ext}} = 0$ , then:

$$\frac{d\vec{P}}{dt} = 0 \implies \vec{P} = \sum m_i \vec{v}_i = \text{constant} \quad (11)$$

2. **Conservation of Angular Momentum:** If net external torque  $\vec{\tau}_{\text{ext}} = \vec{r} \times \vec{F}_{\text{ext}} = 0$ , then:

$$\frac{d\vec{L}}{dt} = 0 \implies \vec{L} = \vec{r} \times \vec{P} = I\vec{\omega} = \text{constant} \quad (12)$$

3. **Conservation of Energy:** In an isolated system with conservative forces:

$$E_{\text{total}} = K + V = \frac{1}{2}mv^2 + V(\vec{r}) = \text{constant} \quad (13)$$

## 4. Non-Inertial Frames of Reference

### 4.1 Inertial vs. Non-Inertial Frames

#### Definition: Frames of Reference

- **Inertial Frame:** A frame of reference moving with zero acceleration (at rest or constant velocity) in which Newton's laws of motion hold strictly.
- **Non-Inertial Frame:** An accelerating or rotating frame of reference in which Newton's laws do not hold directly unless fictitious / pseudo forces are introduced.

### 4.2 Rotating Frames: Centrifugal and Coriolis Forces

Consider a frame  $S'$  rotating with constant angular velocity  $\vec{\omega}$  relative to a stationary inertial frame  $S$ .

#### Transformation Equation for Acceleration

The transformation relation between acceleration in fixed frame  $\vec{a}_S$  and rotating frame  $\vec{a}_{S'}$  is:

$$\vec{a}_S = \vec{a}_{S'} + 2(\vec{\omega} \times \vec{v}') + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad (14)$$

Multiplying by mass  $m$ , the effective force in the rotating frame is:

$$\vec{F}' = m\vec{a}_{S'} = \vec{F}_{\text{real}} - \underbrace{2m(\vec{\omega} \times \vec{v}')}_{\text{Coriolis Force } \vec{F}_{\text{cor}}} - \underbrace{m\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{Centrifugal Force } \vec{F}_{\text{cent}}} \quad (15)$$

#### Properties of Rotating Fictitious Forces:

| Force                    | Vector Expression                                                                                           | Key Characteristics                                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Centrifugal Force</b> | $\vec{F}_{\text{cent}} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r}) = m\omega^2 r_{\perp} \hat{r}$ | Directed radially outward from axis of rotation. Present even if particle is stationary in rotating frame ( $\vec{v}' = 0$ ).                                           |
| <b>Coriolis Force</b>    | $\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$                                                  | Perpendicular to both velocity $\vec{v}'$ and angular velocity $\vec{\omega}$ . Acts only when particle is in motion ( $\vec{v}' \neq 0$ ). Does zero work ( $W = 0$ ). |

#### Physical Effects of Coriolis Force on Earth:

- **Deflection of Moving Objects:** Moving bodies in Northern Hemisphere are deflected to their *right*, while in Southern Hemisphere to their *left* (causes cyclone rotation direction).
- **Eastward Deflection of Freely Falling Body:** A body dropped from height  $h$  at latitude  $\lambda$  experiences an eastward deflection  $\Delta x = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}}$ .
- **Foucault Pendulum:** Demonstrates Earth's rotation directly via precession of its oscillation plane at angular rate  $\Omega = \omega \sin \lambda$ .

**Solved Problem 4: Eastward Deflection Calculation**

**Problem:** Calculate the eastward deflection due to Coriolis force for a body dropped from a height  $h = 100$  m at Kolkata (latitude  $\lambda = 22.5^\circ\text{N}$ ). (Take  $\omega = 7.29 \times 10^{-5}$  rad/s,  $g = 9.8$  m/s<sup>2</sup>).

**Solution:**

1. Formula for eastward deflection:

$$d = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}} \quad (16)$$

2. Calculate time of fall:  $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{200}{9.8}} = 4.518$  s.

3. Substitute values:  $d = \frac{1}{3}(7.29 \times 10^{-5})(9.8) \cos(22.5^\circ)(4.518)^3 \cos 22.5^\circ = 0.9239$ ,  $(4.518)^3 = 92.23$ .  $d = 0.333 \times 7.29 \times 10^{-5} \times 9.8 \times 0.9239 \times 92.23 = 0.0195$  m = **1.95** cm.

## 5. Harmonic Oscillations

### 5.1 Damped Harmonic Motion

When an oscillator experiences a viscous damping force proportional to velocity ( $F_{\text{damping}} = -b\dot{x}$ ), the equation of motion is:

$$m\ddot{x} + b\dot{x} + kx = 0 \implies \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0 \quad (17)$$

where  $\gamma = \frac{b}{2m}$  is the damping coefficient and  $\omega_0 = \sqrt{\frac{k}{m}}$  is the natural frequency.

Assuming solution  $x(t) = Ce^{rt}$ , the roots of characteristic equation are  $r = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$ .

#### Three Cases of Damped Motion

1. **Overdamped** ( $\gamma > \omega_0$ ): Non-oscillatory exponential return to equilibrium.

$$x(t) = e^{-\gamma t} \left( Ae^{\sqrt{\gamma^2 - \omega_0^2} t} + Be^{-\sqrt{\gamma^2 - \omega_0^2} t} \right) \quad (18)$$

2. **Critically Damped** ( $\gamma = \omega_0$ ): Fastest non-oscillatory return to equilibrium without overshoot. Used in shock absorbers and deadbeat galvanometers.

$$x(t) = (A + Bt)e^{-\gamma t} \quad (19)$$

3. **Underdamped** ( $\gamma < \omega_0$ ): Oscillatory motion with exponentially decaying amplitude.

$$x(t) = A_0 e^{-\gamma t} \cos(\omega_d t + \phi), \quad \text{where } \omega_d = \sqrt{\omega_0^2 - \gamma^2} \quad (20)$$

**Damping Parameters: Logarithmic Decrement and Quality Factor**

- **Relaxation Time ( $\tau$ ):** Time in which energy drops to  $1/e$  of initial value:  $\tau = \frac{1}{2\gamma} = \frac{m}{b}$ .
- **Logarithmic Decrement ( $\delta$ ):** Natural logarithm of ratio of successive amplitudes one period  $T_d$  apart:

$$\delta = \ln \left( \frac{A_n}{A_{n+1}} \right) = \gamma T_d = \frac{b T_d}{2m} \quad (21)$$

- **Quality Factor ( $Q$ ):** Dimensionless measure of sharpness of oscillation energy retention:

$$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma} = \frac{\omega_0 \tau}{1} = \frac{\pi}{\delta} \quad (22)$$

**5.2 Forced Oscillations and Resonance**

Under a periodic external driving force  $F(t) = F_0 \cos(\omega t)$ , the differential equation is:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f_0 \cos(\omega t) \quad \left( \text{where } f_0 = \frac{F_0}{m} \right) \quad (23)$$

In steady state, the response oscillates at driving frequency  $\omega$ :

$$x(t) = x_0 \cos(\omega t - \phi) \quad (24)$$

where steady-state amplitude  $x_0$  and phase lag  $\phi$  are given by:

$$x_0(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2 \omega^2}}, \quad \tan \phi = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} \quad (25)$$

**Resonance Conditions**

- **Amplitude Resonance Frequency ( $\omega_r$ ):** Frequency at which displacement amplitude  $x_0$  is maximum:

$$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}, \quad x_{0,\max} = \frac{f_0}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} \approx \frac{f_0}{2\gamma\omega_0} \quad (\text{for small } \gamma) \quad (26)$$

- **Velocity Resonance Frequency ( $\omega_v$ ):** Frequency at which velocity amplitude  $v_0 = \omega x_0$  is maximum:

$$\omega_v = \omega_0 \quad (\text{Natural undamped frequency}) \quad (27)$$

- **Phase at Resonance:** At velocity resonance ( $\omega = \omega_0$ ),  $\tan \phi = \infty \implies \phi = 90^\circ$  (force and velocity are in phase).

**6. Rigid Body Dynamics and Moment of Inertia**

## 6.1 Rigid Body Motion and Angular Velocity Vector

A rigid body is a system of particles in which inter-particle distances remain constant under external forces ( $|\vec{r}_i - \vec{r}_j| = \text{constant}$ ).

General motion of a rigid body is a combination of translation of center of mass (CM) and rotation about CM. The angular velocity vector  $\vec{\omega}$  is a vector directed along the instantaneous axis of rotation according to the right-hand rule.

Linear velocity of any particle  $i$  at position  $\vec{r}_i$  relative to an axis is:

$$\vec{v}_i = \vec{\omega} \times \vec{r}_i \quad (28)$$

## 6.2 Moment of Inertia and Parallel/Perpendicular Axes Theorems

### Definition: Moment of Inertia

**Moment of Inertia ( $I$ )** measures the rotational inertia of a body about a given axis. For a continuous mass distribution:

$$I = \sum m_i r_i^2 = \int r^2 dm = mk^2 \quad (29)$$

where  $k$  is the **radius of gyration**.

### Fundamental Theorems of Moment of Inertia

1. **Parallel Axis Theorem:** MI about any axis is equal to MI about a parallel axis through CM plus  $Mh^2$ :

$$I = I_{\text{cm}} + Mh^2 \quad (h = \text{perpendicular distance between parallel axes}) \quad (30)$$

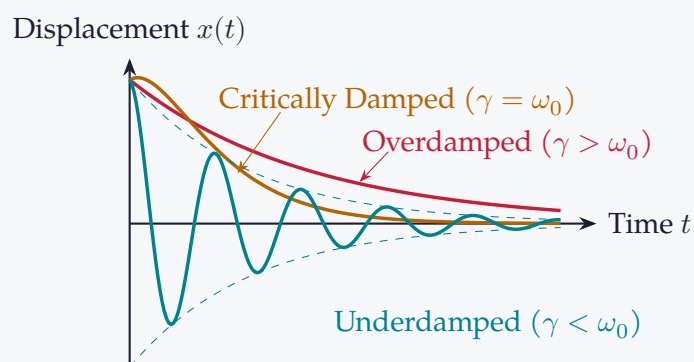
2. **Perpendicular Axis Theorem:** For a planar lamina in  $xy$ -plane, MI about  $z$ -axis perpendicular to lamina is:

$$I_z = I_x + I_y \quad (31)$$

**Standard Moments of Inertia Summary:**

| Body                           | Axis of Rotation                     | Moment of Inertia ( $I$ ) |
|--------------------------------|--------------------------------------|---------------------------|
| Thin Uniform Rod (Length $L$ ) | Perpendicular through center.        | $I = \frac{1}{12}ML^2$    |
| Thin Uniform Rod (Length $L$ ) | Perpendicular through one end.       | $I = \frac{1}{3}ML^2$     |
| Circular Ring (Radius $R$ )    | Central axis perpendicular to plane. | $I = MR^2$                |
| Circular Disc (Radius $R$ )    | Central axis perpendicular to plane. | $I = \frac{1}{2}MR^2$     |
| Solid Sphere (Radius $R$ )     | Any diameter axis.                   | $I = \frac{2}{5}MR^2$     |
| Spherical Shell (Radius $R$ )  | Any diameter axis.                   | $I = \frac{2}{3}MR^2$     |
| Solid Cylinder (Radius $R$ )   | Central longitudinal axis.           | $I = \frac{1}{2}MR^2$     |

### Damped Harmonic Oscillator Response Regimes



## Quick Revision — Module 1

Mechanics | BS-PH101

### Key Formulas and Metrics at a Glance

|                            |                                                                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Degrees of Freedom         | $f = 3N - k$ ( $N$ particles, $k$ constraints)                                                                                                           |
| Limiting Static Friction   | $f_{s,\max} = \mu_s N$ and $\tan \lambda = \mu_s$                                                                                                        |
| Potential Energy Relation  | $\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$ |
| Conservative Force Rule    | $\nabla \times \vec{F} = 0 \iff \oint \vec{F} \cdot d\vec{r} = 0$                                                                                        |
| Coriolis Acceleration      | $\vec{a}_{\text{cor}} = 2(\vec{\omega} \times \vec{v}')$ (Deflection to right in Northern Hemisphere)                                                    |
| Underdamped Frequency      | $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ ( $\gamma = b/2m$ )                                                                                            |
| Quality Factor             | $Q = \frac{\omega_0}{2\gamma} = \frac{\pi}{\delta} = \omega_0 \tau$                                                                                      |
| Resonance Amplitude Freq.  | $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$                                                                                                               |
| Parallel Axis Theorem      | $I = I_{\text{cm}} + Mh^2$                                                                                                                               |
| Perpendicular Axis Theorem | $I_z = I_x + I_y$ (Planar lamina in $xy$ -plane)                                                                                                         |

### One-Line Definitions

- **Holonomic Constraint:** A constraint expressible as an integrable algebraic equation of position coordinates and time  $f(\vec{r}, t) = 0$ .
- **Conservative Force:** A force field in which work done along any closed path is zero ( $\oint \vec{F} \cdot d\vec{r} = 0$ ).
- **Coriolis Force:** A velocity-dependent fictitious force acting in a rotating reference frame ( $\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$ ).
- **Logarithmic Decrement:** The natural log of ratio of two consecutive amplitudes one period apart ( $\delta = \gamma T_d$ ).
- **Radius of Gyration:** The perpendicular distance from axis of rotation where whole mass can be assumed concentrated ( $I = Mk^2$ ).

### Frequently Asked / Viva Questions

**Q1. State the condition for a force field to be conservative.**

A force field  $\vec{F}$  is conservative if  $\nabla \times \vec{F} = 0$  (irrotational) or equivalently work done around any closed loop  $\oint \vec{F} \cdot d\vec{r} = 0$ .

**Q2. Why is Coriolis force called a fictitious force?**

## Frequently Asked / Viva Questions (contd.)

Because it arises due to the acceleration of the observer's rotating frame of reference, not due to physical interactions between bodies.

**Q3. What is the physical significance of negative gradient of potential energy?**

$\vec{F} = -\nabla V$  indicates force always points in direction of steepest decrease of potential energy towards stable equilibrium.

**Q4. Distinguish between holonomic and non-holonomic constraints with examples.**

Holonomic constraints are integrable algebraic position relations (e.g. rigid body length); non-holonomic involve non-integrable velocity relations or inequalities (e.g. sphere rolling without slipping).

**Q5. What is critical damping and state its practical application?**

Critical damping occurs when  $\gamma = \omega_0$ . The system returns to equilibrium in shortest time without oscillating. Used in car shock absorbers and deadbeat galvanometers.

**Q6. How does amplitude resonance frequency differ from velocity resonance frequency?**

Amplitude resonance occurs at  $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$  while velocity resonance occurs exactly at natural frequency  $\omega_v = \omega_0$ .

**Q7. What is Quality Factor ( $Q$ ) of an oscillator?**

$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma}$ . Higher  $Q$  implies lower damping and sharper resonance.

**Q8. State Parallel and Perpendicular axes theorems of moment of inertia.**

Parallel axis:  $I = I_{\text{cm}} + Mh^2$ . Perpendicular axis:  $I_z = I_x + I_y$  for planar laminas.

**Q9. Why are track rails laid higher on outer curves?**

To provide necessary centripetal force via component of normal reaction and eliminate reliance on lateral friction.

**Q10. Calculate the moment of inertia of a solid sphere of mass  $M$  and radius  $R$  about a tangent.**

By parallel axis theorem,  $I_{\text{tangent}} = I_{\text{cm}} + MR^2 = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$ .

**Q11. What is the total energy of a simple harmonic oscillator of mass  $m$  and amplitude  $A$ ?**

$E = \frac{1}{2}m\omega_0^2 A^2$ . Total mechanical energy is constant and continuously converts between kinetic and potential forms.

**Q12. Show that Coriolis force does no work on a moving particle.**

$\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$  is always perpendicular to velocity  $\vec{v}'$ . Power  $P = \vec{F}_{\text{cor}} \cdot \vec{v}' = 0 \Rightarrow W = 0$ .

## Summary Comparison of Damped Motion Regimes

| Parameter          | Overdamped<br>( $\gamma > \omega_0$ ) | Critically Damped<br>( $\gamma = \omega_0$ ) | Underdamped<br>( $\gamma < \omega_0$ )            |
|--------------------|---------------------------------------|----------------------------------------------|---------------------------------------------------|
| Roots of Char. Eq. | Real and distinct.                    | Real and equal<br>( $r = -\gamma$ ).         | Complex conjugate<br>( $-\gamma \pm i\omega_d$ ). |
| Motion Nature      | Non-oscillatory, slow return.         | Non-oscillatory, fastest return.             | Oscillatory with decaying amplitude.              |
| Frequency          | No frequency of oscillation.          | No frequency of oscillation.                 | $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ .       |
| Practical Use      | Heavy door closers.                   | Galvanometers, vehicle suspension.           | Tuning circuits, musical strings.                 |

# Module 2

## Interference vs. Diffraction and Classification

### 1. Interference vs. Diffraction and Classification

#### 1.1 Distinction Between Interference and Diffraction

##### Definition: Interference and Diffraction

- **Interference:** Modification in intensity resulting from the superposition of two or more coherent light waves originating from distinct coherent sources.
- **Diffraction:** The bending of light waves around the sharp edges of an obstacle or aperture and their subsequent superposition within the geometrical shadow region.

##### Comprehensive Comparison:

| Property     | Interference                                                   | Diffraction                                                                                         |
|--------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Wave Origin  | Superposition of waves from two distinct coherent sources.     | Superposition of secondary wavelets originating from different points of the <i>same</i> wavefront. |
| Fringe Width | Fringes are usually of equal width ( $\beta = \lambda D/d$ ).  | Fringes are never of equal width (central max is twice as wide as secondary max).                   |
| Intensity    | All bright fringes have equal intensity ( $I_{\max} = 4I_0$ ). | Intensity decreases rapidly with increasing order of secondary maxima.                              |
| Darkness     | Minima are completely dark ( $I_{\min} = 0$ ).                 | Minima are generally not completely dark.                                                           |

## 1.2 Fraunhofer vs. Fresnel Diffraction

| Feature   | Fresnel Diffraction                                      | Fraunhofer Diffraction                                   |
|-----------|----------------------------------------------------------|----------------------------------------------------------|
| Distance  | Source and screen are at finite distances from aperture. | Source and screen are at effectively infinite distances. |
| Wavefront | Incident wavefront is spherical or cylindrical.          | Incident wavefront is plane (parallel rays).             |
| Lenses    | No lenses required to focus pattern.                     | Convex lenses required before and after aperture.        |
| Analysis  | Analyzed using Fresnel half-period zones.                | Analyzed using simple wave phase summation.              |

## 2. Fraunhofer Diffraction at Single, Double, and Multiple Slits

### 2.1 Single Slit Diffraction

Consider a plane wavefront of wavelength  $\lambda$  incident normally on a slit of width  $e$ .

#### Single Slit Intensity Formula

The resultant intensity on a screen at diffraction angle  $\theta$  is:

$$I(\theta) = I_0 \left( \frac{\sin \beta}{\beta} \right)^2 \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda} \quad (32)$$

#### Key Conditions:

- **Central Maximum:** At  $\theta = 0 \Rightarrow \beta = 0, \lim_{\beta \rightarrow 0} \frac{\sin \beta}{\beta} = 1 \Rightarrow I = I_0$ .
- **Minima Condition:**  $\sin \beta = 0$  (for  $\beta \neq 0$ )  $\Rightarrow \beta = m\pi \Rightarrow e \sin \theta = m\lambda$  ( $m = \pm 1, \pm 2, \pm 3, \dots$ ).
- **Secondary Maxima:** Occur near  $\beta = \pm 1.43\pi, \pm 2.46\pi, \dots$  with intensity ratios:

$$I_0 : I_1 : I_2 = 1 : \frac{1}{22} : \frac{1}{61} = 1 : 0.045 : 0.016 \quad (33)$$

- **Width of Central Maximum:**  $\Delta\theta = \frac{2\lambda}{e} \Rightarrow$  Linear Width  $W = \frac{2\lambda f}{e}$  (where  $f$  is lens focal length).

### 2.2 Double Slit Diffraction

For two parallel slits, each of width  $e$  and separated by opaque spacing  $d$  (grating element  $= e + d$ ):

**Double Slit Intensity Formula**

$$I(\theta) = 4I_0 \left( \frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda}, \quad \alpha = \frac{\pi(e+d) \sin \theta}{\lambda} \quad (34)$$

It is the product of single-slit diffraction factor  $\left( \frac{\sin \beta}{\beta} \right)^2$  and interference factor  $\cos^2 \alpha$ .

**Missing Orders (Absent Spectra):** Occurs when an interference maximum coincides with a diffraction minimum:

$$(e+d) \sin \theta = n\lambda \quad \text{and} \quad e \sin \theta = m\lambda \implies \frac{e+d}{e} = \frac{n}{m} \quad (35)$$

If  $e = d$ , then  $\frac{n}{m} = 2 \implies n = 2m$ , so 2<sup>nd</sup>, 4<sup>th</sup>, 6<sup>th</sup> interference orders disappear.

**2.3 Diffraction Grating and Resolving Power**

A plane transmission diffraction grating consists of  $N$  parallel, equally spaced slits with grating element  $(e+d)$ .

**Grating Equations and Resolving Power**

- Principal Maxima Condition:**

$$(e+d) \sin \theta = m\lambda \quad (m = 0, 1, 2, \dots) \quad (36)$$

- Dispersive Power ( $D$ ):** Rate of change of diffraction angle with wavelength:

$$D = \frac{d\theta}{d\lambda} = \frac{m}{(e+d) \cos \theta} \quad (37)$$

- Resolving Power ( $R$ ):** Capacity to separate two spectral lines of wavelength  $\lambda$  and  $\lambda + d\lambda$ :

$$R = \frac{\lambda}{d\lambda} = mN \quad (N = \text{total illuminated rulings}) \quad (38)$$

**Solved Problem 1: Grating Spectral Lines and Maximum Order**

**Problem:** A grating has 5000 lines/cm. Light of wavelength  $\lambda = 6000 \text{ \AA}$  is incident normally. Find: (a) Grating element  $(e+d)$ , (b) Highest observable order.

**Solution:**

1. Grating element  $(e+d) = \frac{1 \text{ cm}}{5000} = 2 \times 10^{-4} \text{ cm} = 2 \times 10^{-6} \text{ m}$ .

2. Maximum angle  $\theta_{\max} = 90^\circ \implies \sin 90^\circ = 1$ .

3. Substitute in grating equation  $(e+d) \sin 90^\circ = m_{\max} \lambda$ :

$$m_{\max} = \frac{(e+d)}{\lambda} = \frac{2 \times 10^{-6} \text{ m}}{600 \times 10^{-9} \text{ m}} = 3.33 \implies m_{\max} = 3 \quad (39)$$

Hence, maximum 3<sup>rd</sup> order spectrum can be observed.

**Solved Problem 2: Single Slit Central Width**

**Problem:** Light of wavelength  $\lambda = 5890 \text{ \AA}$  is incident normally on a slit of width  $0.1 \text{ mm}$ . A convex lens of focal length  $f = 1.0 \text{ m}$  is placed behind the slit. Calculate linear width of central maximum.

**Solution:**

Linear width of central maximum  $W = \frac{2\lambda f}{e}$ .  $W = \frac{2 \times 5890 \times 10^{-10} \text{ m} \times 1.0 \text{ m}}{0.1 \times 10^{-3} \text{ m}} = 11.78 \times 10^{-3} \text{ m} = 11.78 \text{ mm}$ .

### 3. Polarisation of Light

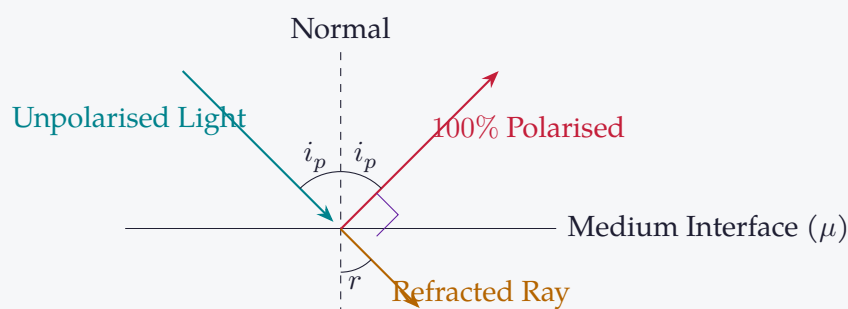
#### 3.1 Brewster's Law and Double Refraction

##### Brewster's Law

When unpolarised light is incident at a specific angle  $i_p$  (polarising angle) on a dielectric interface, the reflected light is 100% plane polarised with electric vector perpendicular to plane of incidence.

$$\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ \quad (40)$$

##### Polarisation by Reflection (Brewster's Law)



**Double Refraction (Birefringence):** When unpolarised light enters an anisotropic crystal (e.g. Calcite, Quartz), it splits into two plane polarised rays travelling at different speeds:

- **Ordinary Ray (O-ray):** Obeys Snell's law; travels with equal velocity in all directions ( $v_o = c/\mu_o$ ).
- **Extraordinary Ray (E-ray):** Violates Snell's law; velocity depends on direction of propagation ( $v_e$ ).

#### 3.2 Retardation Plates: Quarter-Wave and Half-Wave Plates

##### Retardation Plate Thickness Equations

For a crystal plate cut parallel to optic axis of thickness  $t$ :

- **Quarter-Wave Plate (QWP):** Introduces phase difference  $\Delta\phi = \pi/2$  or path

difference  $\Delta x = \lambda/4$ :

$$t = \frac{\lambda}{4|\mu_o - \mu_e|} \quad (41)$$

- **Half-Wave Plate (HWP):** Introduces phase difference  $\Delta\phi = \pi$  or path difference  $\Delta x = \lambda/2$ :

$$t = \frac{\lambda}{2|\mu_o - \mu_e|} \quad (42)$$

### 3.3 Optical Activity and Laurent's Half-Shade Polarimeter

Optically active substances rotate the plane of polarization of linearly polarized light. The angle of rotation  $\theta$  is given by Biot's law:  $\theta = S \cdot L \cdot c$ .

$$\text{Specific Rotation } [S]_{\lambda}^T = \frac{\theta}{L \cdot c} \quad \left( \text{in } \frac{\text{deg}}{\text{dm} \cdot \text{g/cm}^3} \right) \quad (43)$$

where  $\theta$  = rotation angle (deg),  $L$  = path length in decimeters (dm),  $c$  = concentration in  $\text{g/cm}^3$ .

**Laurent's Half-Shade Polarimeter:** Uses a half-shade plate (half quartz, half glass) to divide the field of view into two halves, enabling precise setting of equal brightness.

#### Solved Problem 3: Quarter-Wave Plate Thickness

**Problem:** Calculate the minimum thickness of a quarter-wave plate of Calcite for sodium light of wavelength  $\lambda = 5893 \text{ \AA}$ . Given  $\mu_o = 1.6583$  and  $\mu_e = 1.4864$ .

**Solution:**

1. Calcite is a negative crystal ( $\mu_o > \mu_e$ ), so  $|\mu_o - \mu_e| = 1.6583 - 1.4864 = 0.1719$ .
2. Formula:  $t = \frac{\lambda}{4(\mu_o - \mu_e)}$ .
3. Substitute:  $t = \frac{5893 \times 10^{-10} \text{ m}}{4 \times 0.1719} = \frac{5893 \times 10^{-10}}{0.6876} = 8570 \times 10^{-10} \text{ m} = \mathbf{0.857 \text{ }\mu\text{m}}$ .

## 4. Lasers (Light Amplification by Stimulated Emission of Radiation)

### 4.1 Principles of Laser Action: Einstein's Coefficients

Interaction of radiation with matter involves three fundamental processes:

#### Three Radiation Processes and Einstein's Relations

1. **Stimulated Absorption:** Atom absorbs photon: Rate  $R_{12} = B_{12}N_1u(\nu)$ .
2. **Spontaneous Emission:** Atom decays naturally: Rate  $R_{21,\text{sp}} = A_{21}N_2$ .
3. **Stimulated Emission:** Incident photon induces decay: Rate  $R_{21,\text{st}} = B_{21}N_2u(\nu)$ .

At thermal equilibrium,  $R_{12} = R_{21,\text{sp}} + R_{21,\text{st}}$ . Equating with Planck's law yields Einstein's relations:

$$B_{12} = B_{21} \quad \text{and} \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad (44)$$

## 4.2 Population Inversion and Pumping

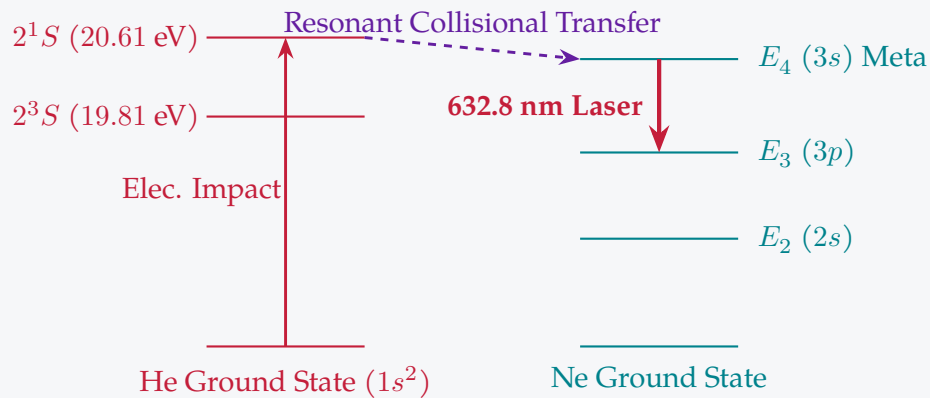
Under normal thermal equilibrium (Boltzmann distribution  $N_2 = N_1 e^{-h\nu/kT}$ ),  $N_1 \gg N_2$ .

### Definition: Population Inversion and Pumping

- **Population Inversion:** The non-equilibrium state in which higher energy level population exceeds lower energy level ( $N_2 > N_1$ ). Necessary condition for optical amplification.
- **Pumping:** The external process of supplying energy to raise atoms to higher excited levels to achieve population inversion (e.g. Optical pumping, Electrical discharge, Direct electron excitation).
- **Metastable State:** An excited state with a long lifetime ( $\sim 10^{-3}$  s compared to normal  $10^{-8}$  s), allowing accumulation of atoms for population inversion.

## 4.3 He-Ne Laser (Gas Laser)

### He-Ne Laser Energy Level Diagram



### Working Mechanism of He-Ne Laser:

1. Electric discharge accelerates electrons which excite He atoms to metastable  $2^1S$  (20.61 eV) state.
2. He atoms transfer energy to Ne atoms via resonant inelastic collisions, populating Ne's  $3s$  ( $E_4$ ) level.
3. Stimulated emission from  $3s \rightarrow 2p$  ( $E_4 \rightarrow E_3$ ) produces red laser light at  $\lambda = 632.8$  nm.

## Quick Revision — Module 2

Optics | BS-PH101

### Key Formulas and Metrics at a Glance

|                                |                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Single Slit Minima             | $e \sin \theta = m\lambda \quad (m = \pm 1, \pm 2, \dots)$                                                      |
| Single Slit Central Width      | $\Delta\theta = \frac{2\lambda}{e} \implies \text{Linear Width } W = \frac{2\lambda f}{e}$                      |
| Double Slit Missing Orders     | $\frac{e+d}{e} = \frac{n}{m} \quad (n = \text{interf. order}, m = \text{diff. order})$                          |
| Grating Equation               | $(e + d) \sin \theta = m\lambda$                                                                                |
| Grating Dispersive Power       | $D = \frac{d\theta}{d\lambda} = \frac{m}{(e+d) \cos \theta}$                                                    |
| Grating Resolving Power        | $R = \frac{\lambda}{d\lambda} = mN$                                                                             |
| Brewster's Law                 | $\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ$                                                      |
| Quarter-Wave Plate (QWP)       | $t = \frac{\lambda}{4 \mu_o - \mu_e }$                                                                          |
| Half-Wave Plate (HWP)          | $t = \frac{\lambda}{2 \mu_o - \mu_e }$                                                                          |
| Specific Rotation              | $[S]_{\lambda}^T = \frac{\theta}{L \cdot c} \quad (\text{deg} \cdot \text{dm}^{-1} \cdot (\text{g/cm}^3)^{-1})$ |
| Einstein Coefficients Relation | $\frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3} \quad \text{and} \quad B_{12} = B_{21}$                       |

### One-Line Definitions

- **Diffraction Grating:** An optical component with a periodic structure of  $N$  slits that diffracts light into several beams.
- **Brewster Angle ( $i_p$ ):** Angle of incidence at which light with a particular polarization is perfectly transmitted without reflection.
- **Birefringence:** The optical property of a material having a refractive index that depends on the polarization and propagation direction of light.
- **Population Inversion:** State in which a system exists with more members in an excited state than in lower energy states ( $N_2 > N_1$ ).
- **Metastable State:** An excited atomic state with a long lifetime ( $\sim 10^{-3}$  s), essential for building up population inversion.

### Frequently Asked / Viva Questions

#### Q1. What is the basic difference between interference and diffraction?

Interference results from superposition of waves from two distinct coherent sources; diffraction results from superposition of wavelets from different parts of the same wavefront.

## Frequently Asked / Viva Questions (contd.)

**Q2. Why is central maximum in single-slit diffraction twice as wide as secondary maxima?**

Because central maximum extends between first minima on either side ( $\sin \theta = \pm \lambda/e$ ), spanning angular width  $2\lambda/e$ , whereas secondary maxima span width  $\lambda/e$ .

**Q3. What is meant by missing orders in double slit diffraction?**

Missing orders occur when an interference maximum coincides with a diffraction minimum at the same angle  $\theta$ , making that order absent ( $\frac{e+d}{e} = \frac{n}{m}$ ).

**Q4. State Rayleigh's criterion for spectral resolution.**

Two spectral lines are just resolved when the central principal maximum of one falls on the first minimum of the other.

**Q5. How does resolving power of a grating differ from dispersive power?**

Dispersive power ( $d\theta/d\lambda$ ) measures angular separation between lines; resolving power ( $\lambda/d\lambda = mN$ ) measures ability to distinguish two close wavelengths.

**Q6. State Brewster's law and show reflected and refracted rays are perpendicular.**

$\mu = \tan i_p$ . Since  $\frac{\sin i_p}{\sin r} = \tan i_p = \frac{\sin i_p}{\cos i_p} \Rightarrow \sin r = \cos i_p = \sin(90^\circ - i_p) \Rightarrow i_p + r = 90^\circ$ .

**Q7. What are O-ray and E-ray in a birefringent crystal?**

O-ray (Ordinary) obeys Snell's law with constant speed  $v_o$ ; E-ray (Extraordinary) violates Snell's law with directional speed  $v_e$ .

**Q8. Distinguish between positive and negative uniaxial crystals.**

In positive crystals ( $\mu_e > \mu_o$ , e.g. Quartz),  $v_o > v_e$ ; in negative crystals ( $\mu_o > \mu_e$ , e.g. Calcite),  $v_e > v_o$ .

**Q9. Explain the function of a Quarter-Wave Plate (QWP).**

A QWP introduces a path difference of  $\lambda/4$  ( $\pi/2$  phase shift) between O-ray and E-ray, converting linearly polarized light to circularly or elliptically polarized light.

**Q10. What is population inversion and why is it necessary for laser action?**

Population inversion ( $N_2 > N_1$ ) ensures that rate of stimulated emission exceeds rate of stimulated absorption, producing net optical amplification.

**Q11. Why is Helium gas added to Neon gas in a He-Ne laser?**

He atoms are excited by electron impact to metastable states ( $2^1S, 3^3S$ ) and efficiently transfer energy to Ne atoms via resonant collisions, achieving population inversion in Ne.

**Q12. Derive relation between Einstein A and B coefficients.**

Equating thermal equilibrium rate  $B_{12}N_1u(\nu) = A_{21}N_2 + B_{21}N_2u(\nu)$  with Boltzmann distribution  $N_2/N_1 = e^{-h\nu/kT}$  and Planck's radiation law gives  $B_{12} = B_{21}$  and  $A_{21}/B_{21} = 8\pi h\nu^3/c^3$ .

## Summary Comparison of Laser Processes

| Process                     | Incident Condition                                  | Transition                           | Output Properties                                     |
|-----------------------------|-----------------------------------------------------|--------------------------------------|-------------------------------------------------------|
| <b>Absorption</b>           | Photon<br>( $h\nu = E_2 - E_1$ )                    | Lower $E_1 \rightarrow$ Higher $E_2$ | Atom reaches excited state.                           |
| <b>Spontaneous Emission</b> | Natural decay<br>(lifetime $\sim 10^{-8}\text{s}$ ) | Higher $E_2 \rightarrow$ Lower $E_1$ | Incoherent, random phase/direction.                   |
| <b>Stimulated Emission</b>  | Triggering photon<br>( $h\nu$ )                     | Higher $E_2 \rightarrow$ Lower $E_1$ | Coherent, identical phase, direction, and wavelength. |

# Module 3

## Maxwell's Equations and Electromagnetic Waves

### 1. Maxwell's Equations and Electromagnetic Waves

#### 1.1 Maxwell's Four Fundamental Equations

Maxwell unified electricity and magnetism into four differential and integral equations governing electromagnetic phenomena.

##### Maxwell's Equations in Differential and Integral Forms

1. **Gauss's Law for Electrostatics:** Net electric flux through a closed surface equals charge enclosed:

$$\nabla \cdot \vec{D} = \rho \iff \oint_S \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}} \quad (\vec{D} = \epsilon_0 \vec{E} + \vec{P}) \quad (45)$$

2. **Gauss's Law for Magnetostatics:** Magnetic monopoles do not exist (magnetic flux lines are continuous loops):

$$\nabla \cdot \vec{B} = 0 \iff \oint_S \vec{B} \cdot d\vec{A} = 0 \quad (46)$$

3. **Faraday's Law of Electromagnetic Induction:** Time-varying magnetic field produces an electric field:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \iff \oint_C \vec{E} \cdot d\vec{r} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{A} \quad (47)$$

4. **Ampere-Maxwell Law:** Magnetic fields are generated by electric currents and time-varying electric fields:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \iff \oint_C \vec{H} \cdot d\vec{r} = \int_S \left( \vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{A} \quad (48)$$

#### 1.2 Displacement Current and Wave Equation in Free Space

Maxwell modified Ampere's law ( $\nabla \times \vec{H} = \vec{J}$ ) by taking the divergence of both sides. Since  $\nabla \cdot (\nabla \times \vec{H}) = 0$ , it required  $\nabla \cdot \vec{J} = 0$ , which violates the charge continuity equation

$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$  for time-varying fields.

Substituting  $\rho = \nabla \cdot \vec{D}$  into continuity equation:

$$\nabla \cdot \vec{J} + \frac{\partial}{\partial t}(\nabla \cdot \vec{D}) = 0 \implies \nabla \cdot \left( \vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = 0 \quad (49)$$

Thus, Maxwell introduced the **Displacement Current Density**  $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$ .

#### Electromagnetic Wave Equation and Speed of Light

In free space ( $\rho = 0, \vec{J} = 0, \mu = \mu_0, \epsilon = \epsilon_0$ ), taking curl of Faraday's law yields:

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B}) \implies \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad (50)$$

Since  $\nabla \cdot \vec{E} = 0$ , this simplifies to the standard 3D wave equation:

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{and} \quad \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \quad (51)$$

Comparing with  $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$ , the wave velocity in vacuum is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}} = 2.998 \times 10^8 \text{ m/s} \quad (52)$$

### 1.3 Poynting Vector and Energy Flow

#### Definition: Poynting Vector

The **Poynting Vector** ( $\vec{S}$ ) represents the directional energy flux density (rate of energy transfer per unit area) of an electromagnetic field:

$$\vec{S} = \vec{E} \times \vec{H} \quad \left( \text{in } \frac{\text{W}}{\text{m}^2} \right) \quad (53)$$

The time-averaged Poynting vector intensity for a plane wave is  $I = \langle S \rangle = \frac{1}{2} c \epsilon_0 E_0^2$ .

#### Solved Problem 1: Displacement Current in a Capacitor

**Problem:** A parallel-plate capacitor with circular plates of radius  $R = 0.1 \text{ m}$  is being charged by a current  $I = 2.0 \text{ A}$ . Calculate: (a) Displacement current  $I_d$ , (b) Rate of change of electric field  $dE/dt$ .

**Solution:**

1. Total displacement current between plates equals conduction current in wires:  
 $I_d = 2.0 \text{ A}$ .
2. Plate area  $A = \pi R^2 = \pi(0.1)^2 = 0.0314 \text{ m}^2$ .

$$3. I_d = \epsilon_0 A \frac{dE}{dt} \Rightarrow \frac{dE}{dt} = \frac{I_d}{\epsilon_0 A}:$$

$$\frac{dE}{dt} = \frac{2.0}{(8.854 \times 10^{-12}) \times 0.0314} = 7.19 \times 10^{11} \text{ V/(m} \cdot \text{s)} \quad (54)$$

## 2. Dielectric Properties of Materials

### 2.1 Dielectric Vectors: Polarisation, Susceptibility, Permittivity

Dielectrics are electrical insulators that can be polarized by an applied electric field.

#### Fundamental Dielectric Relations

- **Electric Dipole Moment ( $\vec{p}$ ):**  $\vec{p} = q\vec{d}$  (directed from  $-q$  to  $+q$ ).
- **Polarisation Vector ( $\vec{P}$ ):** Dipole moment per unit volume:  $\vec{P} = N\vec{p}$ .
- **Electric Displacement ( $\vec{D}$ ):**

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \epsilon_r \vec{E} \quad (55)$$

- **Electric Susceptibility ( $\chi_e$ ):**  $\vec{P} = \epsilon_0 \chi_e \vec{E} \Rightarrow \epsilon_r = 1 + \chi_e$ .

#### Microscopic Polarisation Mechanisms:

| Mechanism                                    | Origin / Description                                                      | Temperature Dependence                                     |
|----------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Electronic (<math>\alpha_e</math>)</b>    | Displacement of electron cloud relative to nucleus in an atom.            | Independent of temperature ( $T$ ).                        |
| <b>Ionic (<math>\alpha_i</math>)</b>         | Relative displacement of positive and negative ions in a crystal lattice. | Independent of temperature ( $T$ ).                        |
| <b>Orientational (<math>\alpha_o</math>)</b> | Realignment of permanent molecular dipoles along external field.          | Strongly temperature dependent ( $\alpha_o \propto 1/T$ ). |
| <b>Space-Charge (<math>\alpha_s</math>)</b>  | Accumulation of free charge carriers at interfaces/grain boundaries.      | Dependent on temperature and frequency.                    |

### 2.2 Clausius-Mossotti Equation

Consider a solid dielectric where local field acting on an atom is the **Lorentz Field**:

$$E_{\text{local}} = E + \frac{P}{3\epsilon_0} \quad (56)$$

The induced dipole moment per atom is  $p = \alpha E_{\text{local}}$ , so total polarisation for  $N$  atoms per unit volume is:

$$P = Np = N\alpha E_{\text{local}} = N\alpha \left( E + \frac{P}{3\epsilon_0} \right) \quad (57)$$

**Clausius-Mossotti Relation Derivation**

Solving for  $P$ :

$$P \left( 1 - \frac{N\alpha}{3\epsilon_0} \right) = N\alpha E \Rightarrow P = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}} \quad (58)$$

Since  $P = (\epsilon_r - 1)\epsilon_0 E$ , substitute for  $P$ :

$$(\epsilon_r - 1)\epsilon_0 E = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}} \Rightarrow \frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0} \quad (59)$$

This connects the macroscopic dielectric constant  $\epsilon_r$  with the microscopic atomic polarizability  $\alpha$ .

### 3. Magnetic Properties of Materials

#### 3.1 Magnetic Vectors and Classification

**Magnetic Field Relations**

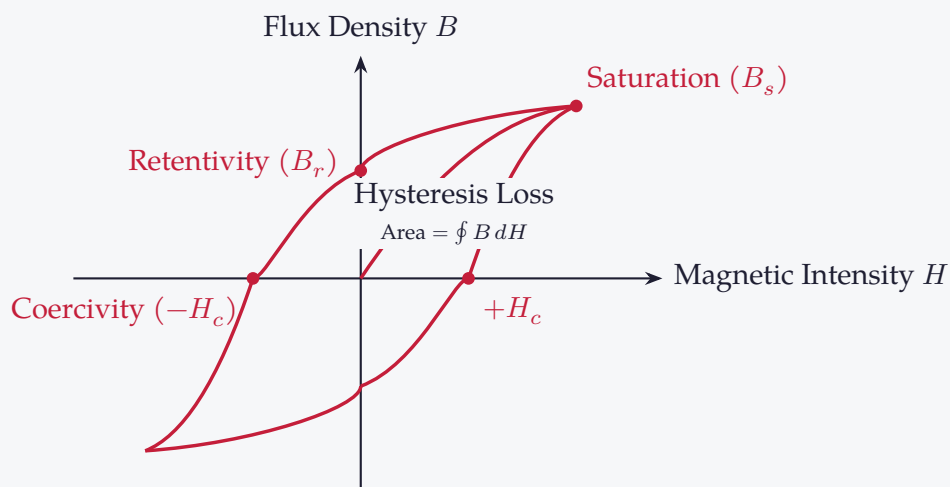
- **Magnetisation ( $\vec{M}$ ):** Magnetic dipole moment per unit volume:  $\vec{M} = \frac{\sum \vec{m}}{V}$ .
- **Magnetic Flux Density ( $\vec{B}$ ):**  $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0\mu_r\vec{H}$ .
- **Magnetic Susceptibility ( $\chi_m$ ):**  $\vec{M} = \chi_m\vec{H} \Rightarrow \mu_r = 1 + \chi_m$ .

#### Classification of Magnetic Materials:

| Property                          | Diamagnetic                                | Paramagnetic                              | Ferromagnetic                                          |
|-----------------------------------|--------------------------------------------|-------------------------------------------|--------------------------------------------------------|
| <b>Atomic Dipoles</b>             | No permanent dipoles.                      | Permanent dipoles present (random).       | Permanent dipoles strongly aligned in domains.         |
| <b>Susceptibility</b><br>$\chi_m$ | Small and negative<br>( $\sim -10^{-5}$ ). | Small and positive<br>( $\sim 10^{-4}$ ). | Very large and positive<br>( $\sim 10^3 - 10^5$ ).     |
| <b>Permittivity</b> $\mu_r$       | $\mu_r < 1$ slightly.                      | $\mu_r > 1$ slightly.                     | $\mu_r \gg 1$ .                                        |
| <b>Temp. Dependence</b>           | Independent of $T$ .                       | Curie Law ( $\chi \propto 1/T$ ).         | Curie-Weiss Law<br>( $\chi \propto \frac{1}{T-T_c}$ ). |
| <b>External Field</b>             | Repelled by magnetic fields.               | Feebly attracted to magnetic fields.      | Strongly attracted into magnetic fields.               |
| <b>Examples</b>                   | Bismuth, Copper, Water, Gold.              | Aluminium, Platinum, Oxygen.              | Iron, Cobalt, Nickel, Gadolinium.                      |

### 3.2 Hysteresis Loop (B-H Curve)

#### Ferromagnetic Hysteresis (B-H) Loop



#### Hysteresis Terminology

- **Retentivity (Remanence  $B_r$ ):** Residual flux density remaining in material when magnetising field  $H$  is reduced to zero.
- **Coercivity ( $H_c$ ):** Reverse magnetising field required to reduce residual flux density  $B$  to zero.
- **Hysteresis Loss:** Energy dissipated per unit volume per cycle as heat during magnetisation reversal, equal to area enclosed by  $B$ - $H$  loop:  $W_{\text{loss}} = \oint B dH$ .

#### Soft vs. Hard Magnetic Materials:

| Feature                | Soft Magnetic Materials                       | Hard Magnetic Materials                                    |
|------------------------|-----------------------------------------------|------------------------------------------------------------|
| <b>Hysteresis Area</b> | Thin and narrow loop (low energy loss).       | Broad and wide loop (high energy loss).                    |
| <b>Coercivity</b>      | Very low coercivity ( $H_c$ small).           | Very high coercivity ( $H_c$ large).                       |
| <b>Retentivity</b>     | High retentivity, easily demagnetised.        | High retentivity, retains magnetisation.                   |
| <b>Applications</b>    | Transformer cores, electromagnets, inductors. | Permanent magnets, magnetic recording tapes, loudspeakers. |

#### Solved Problem 2: Hysteresis Loss Power Dissipation

**Problem:** An iron ring of volume  $V = 100 \text{ cm}^3$  is subjected to a 50 Hz alternating magnetic field. If the area of the hysteresis loop corresponds to  $250 \text{ J/m}^3$  per cycle, calculate the power loss in watts.

**Solution:**

1. Volume  $V = 100 \times 10^{-6} \text{ m}^3 = 10^{-4} \text{ m}^3$ .
2. Frequency  $f = 50 \text{ Hz}$  (50 cycles/second).

3. Energy loss per second (Power  $P$ ) = Energy/cycle/volume  $\times$  Volume  $\times$  Frequency:

$$P = W_{\text{loss}} \times V \times f = 250 \text{ J/m}^3 \times 10^{-4} \text{ m}^3 \times 50 \text{ s}^{-1} = \mathbf{1.25 \text{ W}} \quad (60)$$

## Quick Revision — Module 3

Electromagnetism & Materials | BS-PH101

### Key Formulas and Metrics at a Glance

|                            |                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Maxwell 1 (Gauss Electro)  | $\nabla \cdot \vec{D} = \rho \iff \oint_S \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}}$                                                  |
| Maxwell 2 (Gauss Magneto)  | $\nabla \cdot \vec{B} = 0 \iff \oint_S \vec{B} \cdot d\vec{A} = 0$                                                                       |
| Maxwell 3 (Faraday)        | $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \iff \oint_C \vec{E} \cdot d\vec{r} = -\frac{\partial \Phi_B}{\partial t}$ |
| Maxwell 4 (Ampere-Maxwell) | $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$                                                                  |
| Displacement Current       | $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$                                       |
| Speed of Light in Vacuum   | $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$                                                                      |
| Poynting Vector            | $\vec{S} = \vec{E} \times \vec{H} \quad (\text{W/m}^2)$                                                                                  |
| Clausius-Mossotti Eq.      | $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$                                                                    |
| Magnetic Relations         | $\vec{B} = \mu_0(\vec{H} + \vec{M}) \quad \text{and} \quad \mu_r = 1 + \chi_m$                                                           |
| Hysteresis Power Loss      | $P_{\text{loss}} = (\oint B dH) \times \text{Volume} \times \text{Frequency}$                                                            |

### One-Line Definitions

- **Displacement Current:** A quantity appearing in Maxwell's equations defined by rate of change of electric displacement field ( $\vec{J}_d = \partial \vec{D} / \partial t$ ).
- **Poynting Vector:** A vector representing the directional energy flux density of an electromagnetic field ( $\vec{S} = \vec{E} \times \vec{H}$ ).
- **Lorentz Local Field:** The actual local electric field acting at the position of an individual atom in a solid ( $E_{\text{local}} = E + P/3\epsilon_0$ ).
- **Retentivity ( $B_r$ ):** Residual magnetic flux density remaining in a ferromagnetic material when magnetising field  $H$  is reduced to zero.
- **Coercivity ( $H_c$ ):** The magnitude of reverse magnetic field intensity required to reduce residual magnetisation to zero.

### Frequently Asked / Viva Questions

**Q1. Why did Maxwell modify Ampere's law and what term did he add?**

Ampere's law ( $\nabla \times \vec{H} = \vec{J}$ ) violated charge conservation for time-varying fields ( $\nabla \cdot \vec{J} \neq 0$ ). Maxwell added displacement current density  $\vec{J}_d = \partial \vec{D} / \partial t$ .

**Q2. What is the physical significance of  $\nabla \cdot \vec{B} = 0$ ?**

## Frequently Asked / Viva Questions (contd.)

It signifies that isolated magnetic monopoles do not exist in nature and magnetic flux lines always form continuous closed loops.

**Q3. What is Poynting vector and state its SI unit?**

$\vec{S} = \vec{E} \times \vec{H}$  represents the rate of electromagnetic energy flow per unit area. Its SI unit is  $\text{W/m}^2$ .

**Q4. State the relation between dielectric constant  $\epsilon_r$  and electric susceptibility  $\chi_e$ .**

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0(1 + \chi_e)\vec{E} = \epsilon_0 \epsilon_r \vec{E} \implies \epsilon_r = 1 + \chi_e.$$

**Q5. Derive Clausius-Mossotti relation connecting macroscopic and microscopic dielectric properties.**

Using Lorentz local field  $E_{\text{local}} = E + P/3\epsilon_0$ , total polarisation  $P = N\alpha E_{\text{local}} = N\alpha(E + P/3\epsilon_0)$ . Substituting  $P = (\epsilon_r - 1)\epsilon_0 E$  yields  $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$ .

**Q6. Distinguish between polar and non-polar dielectrics.**

Polar dielectrics have permanent molecular dipole moments (e.g.  $\text{H}_2\text{O}$ ,  $\text{HCl}$ ); non-polar dielectrics have zero permanent dipole moment in zero field (e.g.  $\text{O}_2$ ,  $\text{N}_2$ ,  $\text{CO}_2$ ).

**Q7. Why does orientational polarizability decrease with increasing temperature?**

Because higher thermal agitation randomizes the orientation of permanent molecular dipoles, opposing their alignment along the external field ( $\alpha_o \propto 1/T$ ).

**Q8. Distinguish between diamagnetic and paramagnetic materials.**

Diamagnetic materials have no permanent dipoles, negative susceptibility ( $\chi_m < 0$ ), and are repelled by magnetic fields. Paramagnetic materials have permanent dipoles, positive susceptibility ( $\chi_m > 0$ ), and are feebly attracted.

**Q9. State Curie-Weiss law for ferromagnetic materials.**

Above the Curie temperature  $T_c$ , a ferromagnetic material becomes paramagnetic with susceptibility  $\chi_m = \frac{C}{T - T_c}$ .

**Q10. What is meant by hysteresis loop area?**

The area enclosed by the  $B$ - $H$  loop represents the energy dissipated as heat per unit volume per magnetization reversal cycle ( $W_{\text{loss}} = \oint B dH$ ).

**Q11. Which magnetic material is suitable for transformer cores and why?**

Soft magnetic materials (e.g. Soft Iron, Silicon Steel) because they have low coercivity and thin hysteresis loops, minimizing energy loss per cycle.

**Q12. Show that speed of EM wave in vacuum is  $c = 1/\sqrt{\mu_0 \epsilon_0}$ .**

From Maxwell's free-space wave equation  $\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$ , comparing with standard wave equation  $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$  yields  $v = c = 1/\sqrt{\mu_0 \epsilon_0} \approx 3 \times 10^8 \text{ m/s}$ .

## Summary Comparison of Dielectric Polarization Mechanisms

| Mechanism                           | Displacement Type                       | Frequency Range                | Temperature Dependence                       |
|-------------------------------------|-----------------------------------------|--------------------------------|----------------------------------------------|
| <b>Electronic</b> ( $\alpha_e$ )    | Electron cloud relative to nucleus.     | Optical ( $\sim 10^{15}$ Hz).  | Independent of $T$ .                         |
| <b>Ionic</b> ( $\alpha_i$ )         | Positive and negative ion displacement. | Infrared ( $\sim 10^{13}$ Hz). | Independent of $T$ .                         |
| <b>Orientational</b> ( $\alpha_o$ ) | Alignment of permanent dipoles.         | Microwave ( $\sim 10^9$ Hz).   | Inverse with $T$ ( $\alpha_o \propto 1/T$ ). |
| <b>Space-Charge</b> ( $\alpha_s$ )  | Migration of charge carriers.           | Audio ( $\sim 10^3$ Hz).       | Complex $T$ dependence.                      |

# Module 4

## Inadequacy of Classical Physics and Black Body Radiation

### 1. Inadequacy of Classical Physics and Black Body Radiation

#### 1.1 Failure of Classical Mechanics and Planck's Radiation Law

Classical physics (Rayleigh-Jeans law  $u(\nu)d\nu = \frac{8\pi\nu^2}{c^3}k_B T d\nu$ ) predicted that black body radiation energy density tends to infinity at short wavelengths ( $\lambda \rightarrow 0$ ), known as the **Ultraviolet Catastrophe**.

##### Planck's Quantum Hypothesis (1900)

Black body cavity walls consist of atomic oscillators that absorb or emit electromagnetic radiation in discrete packets called **quanta** (photons) of energy:

$$E = nh\nu \quad (n = 1, 2, 3, \dots) \quad (61)$$

where  $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$  is Planck's constant.

##### Planck's Black Body Radiation Law Derivation

Average energy of an oscillator  $\langle E \rangle = \frac{\sum_{n=0}^{\infty} nh\nu e^{-nh\nu/k_B T}}{\sum_{n=0}^{\infty} e^{-nh\nu/k_B T}} = \frac{h\nu}{e^{h\nu/k_B T} - 1}$ . Multiplying by number of modes per unit volume  $N(\nu)d\nu = \frac{8\pi\nu^2}{c^3}d\nu$ :

$$u(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1} d\nu \iff u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} d\lambda \quad (62)$$

#### Deduction of Limiting Laws:

- **Rayleigh-Jeans Law (Long  $\lambda$ ):** For  $h\nu \ll k_B T$ ,  $e^{h\nu/k_B T} - 1 \approx \frac{h\nu}{k_B T} \implies u(\nu)d\nu = \frac{8\pi\nu^2}{c^3}k_B T d\nu$ .
- **Wien's Distribution Law (Short  $\lambda$ ):** For  $h\nu \gg k_B T$ ,  $e^{h\nu/k_B T} - 1 \approx e^{h\nu/k_B T} \implies u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda k_B T} d\lambda$ .

### 2. Compton Effect, de Broglie Hypothesis, and Matter Waves

## 2.1 Compton Effect

When a high-energy X-ray or  $\gamma$ -ray photon collides elastically with a stationary free electron, the scattered photon emerges with a longer wavelength ( $\lambda' > \lambda$ ).

### Derivation of Compton Shift Formula

Applying conservation of relativistic energy and momentum during collision:

$$\text{Energy Conservation: } h\nu + m_0c^2 = h\nu' + mc^2 \quad (63)$$

$$\text{Momentum Conservation: } \vec{p}_{\text{photon}} = \vec{p}'_{\text{photon}} + \vec{p}_{\text{electron}} \quad (64)$$

Eliminating electron velocity  $v$  using  $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$  yields the **Compton Shift Equation**:

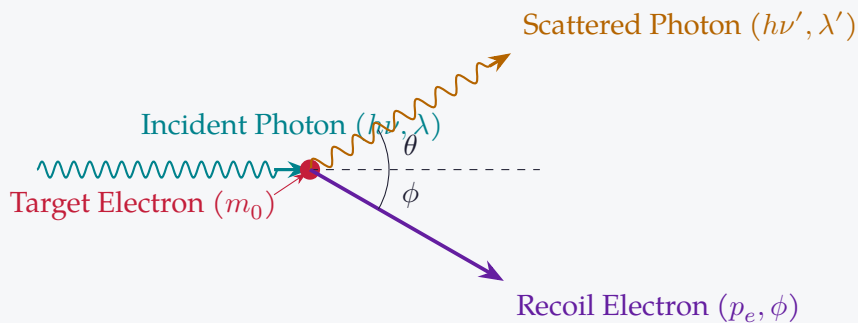
$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0c}(1 - \cos\theta) = 2\lambda_c \sin^2\left(\frac{\theta}{2}\right) \quad (65)$$

where  $\lambda_c = \frac{h}{m_0c} = 0.02426 \text{ \AA} = 2.426 \times 10^{-12} \text{ m}$  is the **Compton Wavelength**.

### Key Characteristics of Compton Shift:

- $\Delta\lambda$  depends *only* on scattering angle  $\theta$ , independent of incident wavelength or target material.
- Maximum shift occurs at  $\theta = 180^\circ$  (backscattering):  $\Delta\lambda_{\text{max}} = \frac{2h}{m_0c} = 0.0485 \text{ \AA}$ .
- Compton shift is absent for visible light because  $\Delta\lambda \ll \lambda_{\text{visible}}$  is unresolvable.

### Compton Scattering Geometry



## 2.2 de Broglie Hypothesis and Davisson-Germer Experiment

**Definition: de Broglie Hypothesis**

Louis de Broglie (1924) proposed that moving matter particles exhibit wave properties with a wavelength ( $\lambda$ ) inversely proportional to momentum ( $p$ ):

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}} \quad (66)$$

For an electron accelerated through potential  $V$  volts:

$$\lambda_e = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad (67)$$

**Davisson-Germer Experiment:** Accelerated electrons at  $V = 54 \text{ V}$  onto a Nickel crystal target observed a pronounced scattering peak at angle  $\phi = 50^\circ$ . Bragg diffraction angle  $\theta = 90^\circ - \phi/2 = 65^\circ$ .

$$\lambda_{\text{Bragg}} = 2d \sin \theta = 2(0.91 \text{ \AA}) \sin 65^\circ = 1.65 \text{ \AA} \quad \text{vs} \quad \lambda_{\text{de Broglie}} = \frac{12.27}{\sqrt{54}} = 1.67 \text{ \AA} \quad (68)$$

This excellent agreement provided direct experimental verification of matter waves.

**Solved Problem 1: Compton Shift at 90 Degrees**

**Problem:** X-rays of wavelength  $\lambda = 1.0 \text{ \AA}$  are scattered by a target at  $\theta = 90^\circ$ . Calculate: (a) Scattered wavelength  $\lambda'$ , (b) Energy of recoil electron.

**Solution:**

1.  $\Delta\lambda = \lambda_c(1 - \cos 90^\circ) = 0.02426 \text{ \AA}(1 - 0) = 0.02426 \text{ \AA}$ .
2. Scattered wavelength  $\lambda' = \lambda + \Delta\lambda = 1.0 + 0.02426 = 1.02426 \text{ \AA}$ .
3. Recoil electron kinetic energy  $K_e = hc \left( \frac{1}{\lambda} - \frac{1}{\lambda'} \right) = hc \frac{\Delta\lambda}{\lambda\lambda'}$ :

$$K_e = \frac{(6.626 \times 10^{-34})(3 \times 10^8)(0.02426 \times 10^{-10})}{(1.0 \times 10^{-10})(1.02426 \times 10^{-10})} = 4.706 \times 10^{-17} \text{ J} = 294.1 \text{ eV} \quad (69)$$

### 3. Heisenberg Uncertainty Principle

#### 3.1 Uncertainty Relations

**Heisenberg Uncertainty Principle Relations**

It is impossible to simultaneously measure both the exact position and exact momentum of a particle with absolute precision:

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} \quad \left( \text{where } \hbar = \frac{h}{2\pi} = 1.054 \times 10^{-34} \text{ J} \cdot \text{s} \right) \quad (70)$$

Alternate canonical conjugate forms:

$$\text{Energy-Time Uncertainty: } \Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad (71)$$

$$\text{Angular Position-Momentum: } \Delta \theta \cdot \Delta L_z \geq \frac{\hbar}{2} \quad (72)$$

### 3.2 Applications: Non-Existence of Electron in Nucleus

#### Proof: Non-Existence of Electron in Nucleus

**Premise:** Radius of atomic nucleus is  $r \approx 5 \times 10^{-15}$  m. If an electron exists inside, maximum position uncertainty  $\Delta x \approx 10^{-14}$  m.

1. Minimum momentum uncertainty:

$$\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{1.054 \times 10^{-34}}{2 \times 10^{-14}} = 5.27 \times 10^{-21} \text{ kg} \cdot \text{m/s} \quad (73)$$

2. Minimum relativistic energy of electron ( $E \approx pc$  for  $p \gg m_0c$ ):

$$E_{\min} \approx \Delta p \cdot c = (5.27 \times 10^{-21}) \times (3 \times 10^8) = 1.58 \times 10^{-12} \text{ J} \approx 9.9 \text{ MeV} \quad (74)$$

3. Experimental  $\beta$ -decay electrons emitted from nucleus have maximum energies of only 2 – 3 MeV.

**Conclusion:** Electrons **cannot reside** inside the nucleus.

## 4. Schrödinger Wave Equation and Wave Function

### 4.1 Physical Interpretation of Wave Function $\psi$

The wave function  $\psi(\vec{r}, t)$  describes the quantum state of a particle.

#### Born's Probability Interpretation of Wave Function

$\psi$  itself is a complex quantity without direct physical measurement. However, the scalar product  $|\psi(\vec{r}, t)|^2 = \psi^* \psi$  represents the **probability density** of finding the particle at position  $\vec{r}$  at time  $t$ .

$$P = \int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = 1 \quad (\text{Normalization Condition}) \quad (75)$$

**Criteria for an Acceptable Quantum Wave Function:**

1.  $\psi$  must be **single-valued** at all points in space.
2.  $\psi$  and its spatial derivative  $\frac{\partial \psi}{\partial x}$  must be **continuous** everywhere.

3.  $\psi$  must be **square-integrable** (normalizable,  $\psi \rightarrow 0$  as  $r \rightarrow \infty$ ).

## 4.2 Time-Dependent and Time-Independent Schrödinger Equations

### Schrödinger Wave Equations

- **1D Time-Dependent Schrödinger Equation:**

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x, t) \psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t} \quad (76)$$

- **1D Time-Independent Schrödinger Equation** (for stationary states  $\psi(x, t) = \psi(x)e^{-iEt/\hbar}$ ):

$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V(x)) \psi(x) = 0 \iff \hat{H}\psi = E\psi \quad (77)$$

where  $\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V$  is the Hamiltonian operator.

## 5. Quantum Applications: Particle in Box, Harmonic Oscillator, H-Atom

### 5.1 Particle in a 1D Infinite Potential Box

Consider a particle of mass  $m$  confined in a 1D box of length  $L$  ( $V(x) = 0$  for  $0 < x < L$ , and  $V(x) = \infty$  outside).

Inside the box ( $0 < x < L$ ),  $\frac{d^2 \psi}{dx^2} + k^2 \psi = 0$  where  $k = \frac{\sqrt{2mE}}{\hbar}$ . Boundary conditions  $\psi(0) = 0$  and  $\psi(L) = 0$  require  $kL = n\pi$ .

### 1D Box Eigenvalues and Normalised Eigenfunctions

$$\text{Energy Eigenvalues: } E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \frac{n^2 h^2}{8mL^2} \quad (n = 1, 2, 3, \dots) \quad (78)$$

$$\text{Normalised Wavefunctions: } \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (79)$$

### Key Consequences:

- **Energy Quantization:** Energy levels are discrete and proportional to  $n^2$ .
- **Zero-Point Energy ( $E_1$ ):** The lowest possible energy is  $E_1 = \frac{h^2}{8mL^2} \neq 0$ , agreeing with uncertainty principle.
- **Degeneracy in 3D Cubic Box:**  $E_{n_x, n_y, n_z} = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)$ . States with different quantum numbers sharing same energy are **degenerate** (e.g.  $(1, 1, 2), (1, 2, 1), (2, 1, 1)$  is 3-fold degenerate).

## 5.2 Quantum Harmonic Oscillator and Hydrogen Atom

### Quantum Harmonic Oscillator and Hydrogen Atom Energies

- **Quantum Harmonic Oscillator** ( $V = \frac{1}{2}m\omega^2x^2$ ):

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega \quad (n = 0, 1, 2, \dots) \quad (80)$$

Zero-point energy  $E_0 = \frac{1}{2}\hbar\omega$  (unlike classical oscillator which can rest at zero energy).

- **Hydrogen Atom (Coulomb Potential  $V = -\frac{e^2}{4\pi\epsilon_0 r}$ ):**

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2n^2} = -\frac{13.6}{n^2} \text{ eV} \quad (n = 1, 2, 3, \dots) \quad (81)$$

Described by four quantum numbers: Principal ( $n$ ), Orbital ( $l$ ), Magnetic ( $m_l$ ), and Spin ( $m_s$ ).

### Solved Problem 2: Particle in 1D Box Transition Energy

**Problem:** An electron is trapped in a 1D infinite box of width  $L = 1.0 \text{ \AA}$ . Calculate: (a) Ground state energy  $E_1$  in eV, (b) Wavelength of photon emitted during transition from  $n = 2 \rightarrow n = 1$ .

**Solution:**

1. Ground state energy  $E_1 = \frac{h^2}{8mL^2}$ :

$$E_1 = \frac{(6.626 \times 10^{-34})^2}{8 \times (9.11 \times 10^{-31}) \times (1.0 \times 10^{-10})^2} = 6.025 \times 10^{-18} \text{ J} = \mathbf{37.66 \text{ eV}} \quad (82)$$

2. First excited energy  $E_2 = 2^2 E_1 = 4(37.66) = 150.64 \text{ eV}$ .

3. Energy difference  $\Delta E = E_2 - E_1 = 3E_1 = 112.98 \text{ eV} = 1.808 \times 10^{-17} \text{ J}$ .

4. Emitted wavelength  $\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{1.808 \times 10^{-17}} = 1.10 \times 10^{-8} \text{ m} = \mathbf{110 \text{ \AA}}$ .

## Quick Revision — Module 4

Quantum Mechanics | BS-PH101

### Key Formulas and Metrics at a Glance

|                            |                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Planck Photon Energy       | $E = h\nu = \frac{hc}{\lambda}$                                                                                    |
| Planck Radiation Law       | $u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} d\lambda$                       |
| Compton Shift Equation     | $\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos\theta)$                                            |
| Compton Wavelength         | $\lambda_c = \frac{h}{m_0 c} = 0.02426 \text{ \AA}$                                                                |
| de Broglie Wavelength      | $\lambda = \frac{h}{p} = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad (\text{for electron})$                           |
| Heisenberg Uncertainty     | $\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad \text{and} \quad \Delta E \cdot \Delta t \geq \frac{\hbar}{2}$ |
| 1D Time-Indep. Schrödinger | $\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0$                                                        |
| 1D Box Energy Eigenvalue   | $E_n = \frac{n^2 h^2}{8mL^2} \quad (n = 1, 2, 3, \dots)$                                                           |
| 1D Box Wavefunction        | $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$                                                 |
| Harmonic Oscillator Energy | $E_n = (n + 1/2)\hbar\omega \quad (\text{Zero-point } E_0 = \frac{1}{2}\hbar\omega)$                               |
| Hydrogen Atom Energy       | $E_n = -\frac{13.6}{n^2} \text{ eV}$                                                                               |

### One-Line Definitions

- **Compton Effect:** The increase in wavelength of X-ray/ $\gamma$ -ray photons when scattered by weakly bound target electrons.
- **Wave-Particle Duality:** The property of matter and light exhibiting both wave-like and particle-like behaviors depending on experimental setup.
- **Probability Density ( $|\psi|^2$ ):** The probability per unit volume of finding a quantum particle at a specific spatial coordinate.
- **Zero-Point Energy:** The lowest possible non-zero ground-state energy that a quantum mechanical physical system may possess ( $E_1 \neq 0$ ).
- **Degeneracy:** The phenomenon where two or more distinct quantum states share the exact same energy eigenvalue.

### Frequently Asked / Viva Questions

#### Q1. What was Rayleigh-Jeans law and why did it fail?

Rayleigh-Jeans law ( $u(\nu) \propto \nu^2 T$ ) predicted infinite energy density at high frequencies ( $\lambda \rightarrow 0$ ), causing the ultraviolet catastrophe.

## Frequently Asked / Viva Questions (contd.)

**Q2. State Planck's quantum hypothesis.**

Atomic oscillators absorb or emit electromagnetic radiation only in discrete energy quanta  $E = nh\nu$ .

**Q3. Why does Compton effect observe no shift at  $\theta = 0^\circ$  and maximum shift at  $\theta = 180^\circ$ ?**

$\Delta\lambda = \lambda_c(1 - \cos\theta)$ . At  $\theta = 0^\circ$ ,  $\cos 0^\circ = 1 \Rightarrow \Delta\lambda = 0$ . At  $\theta = 180^\circ$ ,  $\cos 180^\circ = -1 \Rightarrow \Delta\lambda = 2\lambda_c = 0.0485 \text{ \AA}$ .

**Q4. Why is Compton effect not observable with visible light?**

Visible light has wavelength  $\sim 5000 \text{ \AA}$ . The maximum Compton shift ( $0.0485 \text{ \AA}$ ) is less than  $0.001\%$  of  $\lambda_{\text{visible}}$  and cannot be resolved.

**Q5. What is de Broglie wavelength of an electron accelerated through  $V$  volts?**

$$\lambda = \frac{h}{\sqrt{2mqV}} = \frac{12.27}{\sqrt{V}} \text{ \AA}.$$

**Q6. Explain the physical significance of Davisson-Germer experiment.**

It demonstrated electron diffraction from a Nickel crystal lattice, proving experimentally that moving material particles possess wave characteristics.

**Q7. Prove using uncertainty principle that electron cannot reside inside atomic nucleus.**

For nuclear radius  $\Delta x \approx 10^{-14} \text{ m}$ ,  $\Delta p \geq \hbar/2\Delta x \Rightarrow E \approx \Delta p \cdot c \approx 9.9 \text{ MeV}$ . Experimental  $\beta$ -decay energies are only  $2 - 3 \text{ MeV}$ , proving electrons cannot reside in nucleus.

**Q8. State Born's physical interpretation of wave function  $\psi$ .**

$\psi$  itself has no direct physical measurement, but  $|\psi(\vec{r}, t)|^2 = \psi^*\psi$  represents the probability density of finding the particle at  $\vec{r}$  at time  $t$ .

**Q9. What are the mandatory conditions for an acceptable wave function?**

$\psi$  must be single-valued, continuous everywhere along with its first spatial derivative, and square-integrable ( $\int |\psi|^2 dV = 1$ ).

**Q10. Why is ground state energy of particle in 1D box non-zero?**

If  $E_1 = 0 \Rightarrow p = 0 \Rightarrow \Delta p = 0$ , requiring  $\Delta x = \infty$  by uncertainty principle, contradicting confinement in box of finite length  $L$ .

**Q11. What is zero-point energy of a quantum harmonic oscillator?**

$E_0 = \frac{1}{2}\hbar\omega$ . Unlike classical oscillators, quantum oscillators retain non-zero vibrational energy even at absolute zero temperature ( $T = 0 \text{ K}$ ).

**Q12. What is meant by degeneracy of energy levels?**

Degeneracy occurs when multiple independent quantum wavefunctions (different sets of quantum numbers) correspond to the exact same energy eigenvalue (e.g.  $(1, 1, 2)$ ,  $(1, 2, 1)$ ,  $(2, 1, 1)$  in a 3D box).

## Summary Comparison of Quantum Models

| System Model        | Potential $V(x)$                                 | Ground State Energy ( $E_1$ or $E_0$ ) | Energy Spacing ( $\Delta E$ )                        |
|---------------------|--------------------------------------------------|----------------------------------------|------------------------------------------------------|
| Particle in 1D Box  | $V = 0$ ( $0 < x < L$ ),<br>$V = \infty$ outside | $E_1 = \frac{h^2}{8mL^2}$              | Non-uniform ( $\propto n^2$ ).                       |
| Harmonic Oscillator | $V = \frac{1}{2}m\omega^2 x^2$                   | $E_0 = \frac{1}{2}\hbar\omega$         | Uniform ( $\Delta E = \hbar\omega$ ).                |
| Hydrogen Atom       | $V = -\frac{e^2}{4\pi\epsilon_0 r}$              | $E_1 = -13.6 \text{ eV}$               | Concentrated near continuum<br>( $\propto -1/n^2$ ). |

# Module 5

## Macrostate, Microstate, Phase Space, and Ensembles

### 1. Macrostate, Microstate, Phase Space, and Ensembles

#### 1.1 Basic Concepts of Statistical Mechanics

Statistical mechanics connects the microscopic behavior of individual atoms/molecules with the macroscopic thermodynamic properties ( $P, V, T, S$ ) of a system.

##### Definitions: Macrostate, Microstate, and Phase Space

- **Macrostate:** The macroscopic state of a system specified by overall thermodynamic variables (e.g.  $(N, V, E)$  or  $(N, V, T)$ ).
- **Microstate:** A specific detailed configuration of the system specifying the exact position ( $\vec{r}_i$ ) and momentum ( $\vec{p}_i$ ) of every individual particle.
- **Phase Space:** A  $6N$ -dimensional conceptual space formed by 3 position coordinates  $(x, y, z)$  and 3 momentum coordinates  $(p_x, p_y, p_z)$  for each of  $N$  particles. The elementary phase cell volume in quantum mechanics is  $h^3$ .
- **Thermodynamic Probability ( $W$ ):** The total number of microstates corresponding to a given macrostate.

##### Boltzmann Entropy Relation

$$S = k_B \ln W \quad (83)$$

where  $k_B = 1.38 \times 10^{-23}$  J/K is Boltzmann's constant, and  $W$  is the thermodynamic probability.

#### 1.2 Statistical Ensembles

An **ensemble** is a large collection of independent, macroscopically identical systems existing in different microstates.

**Three Main Gibbs Ensembles:**

| Ensemble               | Fixed Parameters                                                              | Physical System Example                           |
|------------------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Microcanonical</b>  | Number $N$ , Volume $V$ , Energy $E$ (Rigid, isolated, insulated walls).      | Isolated system with zero heat exchange.          |
| <b>Canonical</b>       | Number $N$ , Volume $V$ , Temperature $T$ (Diathermal walls in heat bath).    | Closed system exchanging heat with a reservoir.   |
| <b>Grand Canonical</b> | Chem. potential $\mu$ , Volume $V$ , Temp. $T$ (Permeable, diathermal walls). | Open system exchanging both energy and particles. |

### Solved Problem 1: Microstates and Entropy

**Problem:** Calculate the number of microstates and thermodynamic probability  $W$  for distributing  $N = 4$  distinguishable particles in 2 equal compartments. Find the entropy of the most probable state.

**Solution:**

- Total number of microstates  $= 2^N = 2^4 = 16$ .
- Macrostates are given by  $(n_1, n_2)$  where  $n_1 + n_2 = 4$ :

- $(4, 0) : W = \frac{4!}{4!0!} = 1$
- $(3, 1) : W = \frac{4!}{3!1!} = 4$
- $(2, 2) : W = \frac{4!}{2!2!} = 6$  (Most Probable Macrostate)
- $(1, 3) : W = 4$
- $(0, 4) : W = 1$

- Entropy of most probable macrostate  $(2, 2)$ :

$$S = k_B \ln(6) = (1.38 \times 10^{-23}) \times 1.7917 = 2.47 \times 10^{-23} \text{ J/K} \quad (84)$$

## 2. Density of States

### 2.1 Derivation of Density of States in 3D

The **Density of States** ( $g(E)dE$ ) represents the number of available quantum states per unit volume in the energy range between  $E$  and  $E + dE$ .

#### 3D Free Particle Density of States Derivation

In momentum space, momentum  $p = \sqrt{2mE}$ . The volume of momentum sphere of radius  $p$  is  $V_p = \frac{4}{3}\pi p^3 = \frac{4}{3}\pi(2mE)^{3/2}$ . The number of spatial-momentum phase cells of volume  $h^3$  inside spatial volume  $V$  is:

$$N(E) = \frac{V \cdot \frac{4}{3}\pi(2mE)^{3/2}}{h^3} = \frac{4\pi V(2m)^{3/2}}{3h^3} E^{3/2} \quad (85)$$

Taking the derivative with respect to  $E$  gives  $g(E) = \frac{dN(E)}{dE}$ :

$$g(E)dE = \frac{4\pi V(2m)^{3/2}}{h^3} E^{1/2} dE \quad (86)$$

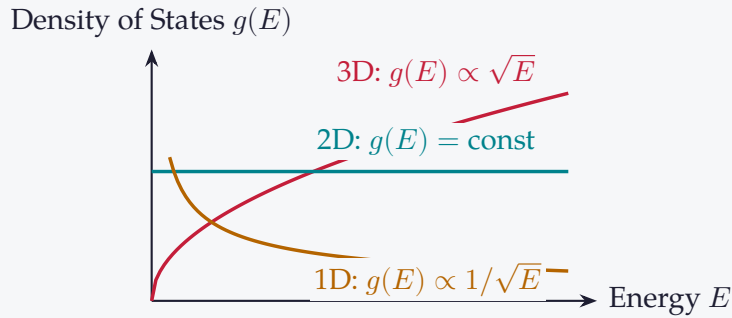
Including electron spin degeneracy  $g_s = 2s + 1 = 2$  for electrons:

$$g_e(E)dE = \frac{8\pi V(2m)^{3/2}}{h^3} E^{1/2} dE \quad (87)$$

### Density of States in Lower Dimensions:

1. **1D System:**  $g_{1D}(E) \propto E^{-1/2}$ .
2. **2D System:**  $g_{2D}(E) = \text{constant}$  (independent of energy  $E$ ).
3. **3D System:**  $g_{3D}(E) \propto E^{1/2}$ .

#### Density of States $g(E)$ in 1D, 2D, and 3D Systems



## 3. Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac Statistics

### 3.1 Qualitative and Quantitative Treatment

Depending on particle distinguishability, quantum spin, and wave function symmetry:

#### Three Distribution Functions

- **Maxwell-Boltzmann (MB):** Classical, distinguishable particles, no spin restriction:

$$f_{\text{MB}}(E) = Ae^{-E/k_B T} \quad (88)$$

- **Bose-Einstein (BE):** Quantum, indistinguishable particles with integer spin ( $s = 0, 1, 2, \dots$ , Bosons):

$$f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1} \quad (89)$$

- **Fermi-Dirac (FD):** Quantum, indistinguishable particles with half-integer spin ( $s = 1/2, 3/2, \dots$ , Fermions), obeys Pauli Exclusion Principle:

$$f_{\text{FD}}(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1} \quad (90)$$

## Comprehensive 8-Criterion Comparison Table:

| Feature              | Maxwell-Boltzmann                                       | Bose-Einstein                                             | Fermi-Dirac                                                    |
|----------------------|---------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| Particle Nature      | Classical,<br>Distinguishable.                          | Quantum,<br>Indistinguishable.                            | Quantum,<br>Indistinguishable.                                 |
| Spin                 | Any spin (Spin<br>ignored).                             | Integral spin $(0, 1\hbar, 2\hbar)$ .                     | Half-integral spin<br>$(\frac{1}{2}\hbar, \frac{3}{2}\hbar)$ . |
| Pauli Principle      | Does not apply.                                         | Does not apply.                                           | Strictly obeys Pauli<br>principle.                             |
| Occupancy/Cell       | Unlimited<br>$(0, 1, 2, \dots, \infty)$ .               | Unlimited<br>$(0, 1, 2, \dots, \infty)$ .                 | At most one particle (0<br>or 1).                              |
| Wave Function        | No wave function<br>overlap.                            | Symmetric under<br>particle exchange.                     | Anti-symmetric under<br>particle exchange.                     |
| Ground State         | All particles rest at<br>$E = 0$ as $T \rightarrow 0$ . | Bose-Einstein<br>Condensation at<br>$T < T_c$ .           | Occupies up to Fermi<br>level $E_F$ at $T = 0$ K.              |
| Physical<br>Examples | Ideal gas molecules<br>(He, Ar).                        | Photons, Phonons,<br>$\alpha$ -particles, $^4\text{He}$ . | Free electrons, Protons,<br>Neutrons, $^3\text{He}$ .          |

## 4. Fermi Energy and Bose-Einstein Condensation

4.1 Fermi Energy and Fermi-Dirac Function at  $T = 0$  K

The **Fermi Energy** ( $E_F$ ) is the highest filled energy level in a fermion system at absolute zero temperature ( $T = 0$  K).

At  $T = 0$  K:

$$f_{\text{FD}}(E) = \begin{cases} 1 & \text{if } E < E_F \\ 0 & \text{if } E > E_F \end{cases} \quad (91)$$

$$\text{Total number of free electrons } N = \int_0^{E_{F0}} g_e(E) dE = \int_0^{E_{F0}} \frac{8\pi V (2m)^{3/2}}{h^3} E^{1/2} dE = \frac{16\pi V (2m)^{3/2}}{3h^3} E_{F0}^{3/2}.$$

Fermi Energy and Average Energy at  $T = 0$  K

Solving for  $E_{F0}$  in terms of electron density  $n = N/V$ :

$$E_{F0} = \frac{h^2}{8m} \left( \frac{3N}{\pi V} \right)^{2/3} = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3} \quad (92)$$

The total kinetic energy of  $N$  electrons at  $T = 0$  K is  $E_{\text{total}} = \int_0^{E_{F0}} E g(E) dE = \frac{3}{5} N E_{F0}$ .  
Average energy per electron at  $T = 0$  K:

$$\langle E \rangle = \frac{3}{5} E_{F0} \quad (93)$$

**Solved Problem 2: Fermi Energy of Copper**

**Problem:** Copper has free electron density  $n = 8.49 \times 10^{28} \text{ m}^{-3}$ . Calculate: (a) Fermi energy  $E_F$  in eV, (b) Fermi velocity  $v_F$ .

**Solution:**

1. Fermi energy  $E_{F0} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$ :  $3\pi^2 n = 3\pi^2(8.49 \times 10^{28}) = 2.514 \times 10^{29} \text{ m}^{-3}$ .  
 $(3\pi^2 n)^{2/3} = 3.983 \times 10^{19} \text{ m}^{-2}$ .

$$E_{F0} = \frac{(1.054 \times 10^{-34})^2}{2 \times (9.11 \times 10^{-31})} \times 3.983 \times 10^{19} = 2.428 \times 10^{-18} \text{ J} = \mathbf{7.04 \text{ eV}} \quad (94)$$

2. Fermi velocity  $v_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2 \times 2.428 \times 10^{-18}}{9.11 \times 10^{-31}}} = \mathbf{1.57 \times 10^6 \text{ m/s}}$ .

**4.2 Bose-Einstein Condensation (BEC)**

At extremely low temperatures below a critical transition temperature  $T_c$ , a substantial fraction of bosons collapse into the lowest quantum ground state ( $E = 0$ ), forming a macroscopic quantum state known as **Bose-Einstein Condensate**.

**Critical Temperature and Ground State Fraction**

Critical BEC transition temperature  $T_c$ :

$$T_c = \frac{h^2}{2\pi m k_B} \left( \frac{N}{2.612 V} \right)^{2/3} \quad (95)$$

Fraction of bosons in ground state for  $T < T_c$ :

$$\frac{N_0}{N} = 1 - \left( \frac{T}{T_c} \right)^{3/2} \quad (96)$$

## Quick Revision — Module 5

Statistical Mechanics | BS-PH101

### Key Formulas and Metrics at a Glance

|                                            |                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Boltzmann Entropy Relation</b>          | $S = k_B \ln W$ ( $W$ = thermodynamic probability)                                              |
| <b>Elementary Phase Cell</b>               | $d\Omega = h^3$ (quantum phase space volume)                                                    |
| <b>3D Density of States</b>                | $g_e(E)dE = \frac{8\pi V(2m)^{3/2}}{h^3} E^{1/2} dE$                                            |
| <b>MB Distribution Function</b>            | $f_{\text{MB}}(E) = Ae^{-E/k_B T}$                                                              |
| <b>BE Distribution Function</b>            | $f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}$ (Bosons, Integer spin)                     |
| <b>FD Distribution Function</b>            | $f_{\text{FD}}(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$ (Fermions, Half-spin)                      |
| <b>Fermi Energy (<math>T = 0</math> K)</b> | $E_{F0} = \frac{h^2}{8m} \left(\frac{3N}{\pi V}\right)^{2/3} = \frac{h^2}{2m} (3\pi^2 n)^{2/3}$ |
| <b>Average Fermi Energy</b>                | $\langle E \rangle = \frac{3}{5} E_{F0}$ (at $T = 0$ K)                                         |
| <b>BEC Transition Temp.</b>                | $T_c = \frac{h^2}{2\pi m k_B} \left(\frac{N}{2.612 V}\right)^{2/3}$                             |
| <b>BEC Condensate Fraction</b>             | $\frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{3/2}$ (for $T < T_c$ )                         |

### One-Line Definitions

- **Microstate:** A specific detailed microscopic specification of position and momentum coordinates for all particles in a system.
- **Ensemble:** A large virtual collection of independent, macroscopically identical systems in different microstates.
- **Fermi Energy ( $E_F$ ):** The maximum energy state occupied by free electrons at absolute zero temperature ( $T = 0$  K).
- **Boson:** A particle with integral quantum spin ( $0, 1\hbar, 2\hbar$ ) obeying Bose-Einstein statistics and symmetric under particle exchange.
- **Bose-Einstein Condensation:** A phase transition where a macroscopic number of bosons collapse into the single lowest energy ground state at  $T < T_c$ .

### Frequently Asked / Viva Questions

#### Q1. What is the difference between a macrostate and a microstate?

A macrostate specifies overall macroscopic variables ( $N, V, E$ ); a microstate specifies exact microscopic position and momentum coordinates ( $\vec{r}_i, \vec{p}_i$ ) of every particle.

#### Q2. What is phase space and what is its minimum volume in quantum mechanics?

## Frequently Asked / Viva Questions (contd.)

Phase space is a  $6N$ -dimensional space combining 3 position and 3 momentum axes per particle. Its minimum volume cell in quantum mechanics is  $h^3$ .

**Q3. Distinguish between Microcanonical, Canonical, and Grand Canonical ensembles.**

Microcanonical keeps  $(N, V, E)$  fixed; Canonical keeps  $(N, V, T)$  fixed (heat exchange allowed); Grand Canonical keeps  $(\mu, V, T)$  fixed (heat and particle exchange allowed).

**Q4. State the basic difference between Bosons and Fermions.**

Bosons have integral spin  $(0, 1\hbar)$ , symmetric wavefunctions, and do not obey Pauli principle. Fermions have half-integral spin  $(\frac{1}{2}\hbar)$ , anti-symmetric wavefunctions, and strictly obey Pauli principle.

**Q5. Why does density of states in 3D vary as  $\sqrt{E}$ ?**

Because the momentum sphere volume  $V_p \propto p^3 \propto E^{3/2}$ . Taking the derivative  $g(E) = dN/dE$  gives  $g(E) \propto E^{1/2} = \sqrt{E}$ .

**Q6. Draw Fermi-Dirac distribution curve at  $T = 0$  K and  $T > 0$  K.**

At  $T = 0$  K,  $f(E)$  is a step function ( $f = 1$  for  $E < E_F$ ,  $f = 0$  for  $E > E_F$ ). At  $T > 0$  K, thermal agitation creates a symmetric tail around  $E_F$  where  $f(E_F) = 1/2$ .

**Q7. What is the physical meaning of Fermi energy?**

Fermi energy represents the highest energy level occupied by free conduction electrons in a metal at absolute zero temperature ( $T = 0$  K).

**Q8. Calculate the average energy of an electron gas at  $T = 0$  K in terms of  $E_F$ .**

$$\langle E \rangle = \frac{\int_0^{E_F} E \cdot E^{1/2} dE}{\int_0^{E_F} E^{1/2} dE} = \frac{\frac{2}{5} E_F^{5/2}}{\frac{2}{3} E_F^{3/2}} = \frac{3}{5} E_F.$$

**Q9. Under what conditions do BE and FD statistics reduce to MB statistics?**

At high temperatures ( $T \rightarrow \infty$ ) and low particle densities ( $n \rightarrow 0$ ), when  $e^{(E-\mu)/k_B T} \gg 1$ , both BE and FD distributions reduce to classical MB distribution  $f(E) \approx e^{-(E-\mu)/k_B T}$ .

**Q10. What is Bose-Einstein Condensation (BEC)?**

BEC is a low-temperature phenomenon where a macroscopic fraction of bosons condense into the single zero-momentum ground state at  $T < T_c$ .

**Q11. Why can photons undergo BEC without fixing total particle number  $N$ ?**

Photons have chemical potential  $\mu = 0$  because photon number is not conserved (photons are freely created/absorbed by cavity walls).

**Q12. Why do electrons in a metal not contribute significantly to heat capacity at room temperature?**

Because Pauli exclusion principle restricts thermal excitation ( $k_B T$ ) to only a small fraction ( $\sim k_B T / E_F \approx 1\%$ ) of electrons near the Fermi level  $E_F$ .

## Summary Comparison of Quantum Statistics

| Property               | Maxwell-Boltzmann              | Bose-Einstein                              | Fermi-Dirac                              |
|------------------------|--------------------------------|--------------------------------------------|------------------------------------------|
| Distribution Eq.       | $f(E) \propto e^{-E/k_B T}$    | $f(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}$   | $f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$ |
| Ground State Occupancy | Non-zero classical energy.     | Condensate ( $N_0$ particles at $E = 0$ ). | Filled Fermi sphere up to $E_F$ .        |
| Wave Symmetry          | Distinguishable (No symmetry). | Symmetric under exchange.                  | Anti-symmetric under exchange.           |